BSc Engineering Sciences – A. Y. 2017/18

Written exam of the course Mathematical Analysis 2

August 30, 2018

Last name: .... First name: ....

Matriculation: ....

Solve the following problems, motivating in detail the answers.

1. Given the power series

$$\sum_{n=1}^{+\infty} \frac{1}{n}\left(1+\frac{1}{n}\right)^{n^2} x^n,$$

determine its radius of convergence r, and study the convergence for $x = \pm r$.

Solution.

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2.

(1) Find all the stationary points of the following scalar field, defined on $\mathbb{R}^2$,

$$f(x, y) = x^2 - 2xy - y + y^3$$

and classify them into relative minima, maxima and saddle points.

(2) Compute the gradient of the following function of (x, y):

$$g(x, y) = xe^{xe^y}.$$

Solution.

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3. Let $\mathbb{f}(x,y) = (e^x, e^y)$. Compute the line integral $\int_C \mathbb{f} \cdot d\boldsymbol{\alpha}$, where C is the segment of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$.

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4. Compute the integral

$$\iint_T dxdy\, x \log(1+\sqrt{x^2+y^2})\ ,$$

where $T = \{(x,y) \in \mathbb{R}^2\ :\ 1 \le x^2+y^2 \le 4,\ x \ge 0\}$.
Solution.

Matriculation: ....

5. Let $\mathbb{F}(x, y, z) = (e^{zy^2},\ e^{yx^2},\ e^{xz^2})$ be a vector field on $\mathbb{R}^3$, C be the boundary of the square with vertices $(0,0,0), (1,0,0), (1,1,0), (0,1,0)$.

Compute the line integral

$$\int_C \mathbb{F} \cdot d\boldsymbol{\alpha},$$

where $\boldsymbol{\alpha}$ is a parametrization of C going counterclockwise.

Solution.